OFFICIAL FACTORY MANUAL • NOVEMBER 2026 EDITION

Complete Operational, Financial & Anti-Theft Handbook

The Mobile-First Operating System for Nigerian Dual-Line Water Plants

> **Executive Guarantee to Mr. Obi:** Know your exact net cash profit from both your Sachet Line and Bottled Water Line before 8:00 PM every evening. Eliminate preform smuggling with digital scale audits, stop driver theft with physical pouches & jar deposits, prevent fake bank transfers with POS RRN locks, and survive power outages with offline-first data caching.

1. Dual-Line Plant Economics & Pricing Architecture

Nigerian water factories operating both sachet pure water and automated PET bottle blow-molding achieve 3.4x higher gross margins than sachet-only plants:

Production Line	Product Unit	Wholesale Price	Unit Margin	Primary Money Leak / Fraud Vulnerability
Line 1: Sachet Pure Water	Bag of 20 Sachets	₦350 wholesale	₦25 – ₦40	Drivers claiming road bursts without returning physical pouches; unrecorded machine rolls.
Line 2: 75cl Bottled Water	Pack of 12 Bottles	₦1,200 wholesale	₦220 – ₦320	PET preforms reported as "blown rejects" and smuggled out for cosmetic packagers.
Line 2: 50cl Bottled Water	Pack of 20 Bottles	₦1,100 wholesale	₦200 – ₦280	Blow molding heat rejects; unrecorded cashier gate sales pocketed as cash.
Line 2: 19L Dispenser Jars	19-Liter Refill	₦900 refill	₦450 – ₦600	**Unreturned or battered empty jars!** Jars cost ₦4,500 – ₦6,000. Driver brings back off-brand cracked jars.

2. Today's Dual-Line Daily Financial Audit (Sample Full Shift)

Revenue Source	Dispatched Volume	Unit Price	Gross Revenue
Sachet Water Line	4,200 Bags (14 Nylon Rolls used)	₦350 / bag	₦1,470,000
75cl Bottled Water Line	200 Shrink Packs (12s)	₦1,200 / pack	₦240,000
50cl Bottled Water Line	150 Shrink Packs (20s)	₦1,100 / pack	₦165,000
19L Dispenser Jar Refills	106 Jars Refilled	₦900 / refill	₦96,000
TOTAL SHIFT GROSS FACTORY REVENUE			**₦1,971,000**

Shift Direct OPEX Breakdown

- **Generator Diesel Overhead:** 32 Liters @ ₦1,350/L = **₦43,200** (Sachet energy overhead = ₦8.20/bag).
- **Sachet Nylon Film:** 14 Rolls (280 kg LDPE) = **₦280,000**.
- **PET Preforms & Caps:** 5,400 units + 24 blow rejects = **₦95,600**.
- **Shrink Wrap Film & Labels:** 350 packs = **₦28,000**.
- **Factory Loaders & Packers Wages:** ₦50,000.
- **Driver Commissions & Fuel Allowance:** ₦50,000.

💰 TODAY'S NET CASH PROFIT: ₦524,200

Physical Cash in Blessing's Drawer: ₦385,000 • Verified Moniepoint Transfers (RRN Locked): ₦1,485,000 • Distributor Credit: ₦65,000.

3. The 4 Zero-Password Staff Roles

Factory staff do not use complex emails or passwords. They tap their picture card and enter their 4-digit PIN:

👑 Mr. Obi (Managing Director & Owner)

PIN: 2026

Access from phone home screen. Audits net cash profit, driver short debits, diesel cost per bag, preform grammage scale audits, and 1-tap WhatsApp evening digest.

📋 Kenneth (Dual-Line Supervisor)

PIN: 1111

Taps "+1 Roll Used" for sachets, logs bottle preforms and blow rejects, verifies scrap scale weight, and dispatches WhatsApp shift close at 6:00 PM.

💰 Blessing (Gate Cashier)

PIN: 2222

Sells sachets, 75cl, 50cl packs, and 19L refills. Logs POS RRN last 4 digits to generate a 3-digit Gate Clearance Pass (GP-xxx) for gate security.

🚚 Musa (Route Driver)

PIN: 3333

Inspects loaded truck manifest, delivers to depots, returns burst plastic pouches, and collects ₦4,500 deposit for unreturned or battered 19L jars.

4. The 5 Hardened Security & Anti-Theft Protocols

1. Server-Side Constant-Time PIN Authentication

> **Zero Client-Side PIN Inspection:** PINs are no longer checked inside client browser JavaScript where tech-savvy workers could view devtools. PIN verification routes through `/api/factory/auth` using constant-time byte comparison and a 5-attempt brute-force rate limiter that locks the terminal for 15 minutes upon repeated failures.

2. Offline-First Persistence Ledger (Zero Data Loss)

In Nigeria, factory phones run out of battery, browsers are accidentally closed, and NEPA drops without notice. The `offlineLedger` automatically caches all rolls, bottle packs, dispenser jars, cashier cash counts, and driver manifests to device LocalStorage with every touch. Reloading restores the shift state instantly.

3. PET Preform Grammage Scrap Scale Verification

> **Stopping Preform Smuggling:** A standard 75cl PET preform weighs 18.0 grams. When operators claim 24 bottles were ruined during blow-molding, Kenneth must place the crushed plastic scrap on a physical digital scale:
>
> **24 Rejects × 18.0g = 432 grams target scrap weight.**
> If the scale weighs 200g, 232g of preforms (13 preforms) were smuggled out to cosmetics or juice packagers. The system displays a red deficit alert until verified.

4. Moniepoint POS Transfer Reference (RRN) Lock & Gate Pass

> **Stopping Fake Bank Alerts:** Cashiers can no longer mark an order as "Transfer" and pocket physical cash. For every bank transfer sale, Blessing must type the last 4 digits of the Moniepoint POS Terminal Reference (RRN). The software immediately prints a 3-digit Gate Clearance Pass (e.g. `GP-842`). Factory gate security will not open the exit boom gate without this code.

5. 19L Dispenser Jar Quality Inspection & Battered Jar Liability

A new 19L polycarbonate dispenser jar costs ₦4,500 – ₦6,000. Drivers often deliver new Obi-branded jars and return cracked or off-brand competitor containers. Under the new protocol, gate loaders inspect returned empty jars:

- Clean Obi Branded Jars: Accepted and credited.
- Battered / Cracked / Third-Party Jars: Rejected at gate. **₦4,500 replacement penalty** is automatically debited from the driver's payout ledger.

5. NAFDAC Laboratory Quality Log (NIS 306:2008 Standard)

Test Parameter	NAFDAC Standard	Obi Factory Safe Range	Status
Total Dissolved Solids (TDS)	< 100 ppm	15 – 45 ppm (Pure Permeate)	✅ Pass (24 ppm)
Water pH Level	6.5 – 7.5	6.8 – 7.4 (Neutral)	✅ Pass (7.2 pH)
UV Sterilizer Lamp Hours	< 8,000 hrs	Active Lamp Lifetime	✅ 2,140 hrs (Safe)
Physical Taste & Clarity	Odorless, Crystal Clear	100% Inspection	✅ Verified Pure

6. Technical Support & Factory Contact

> **Bethelmind Digital Solutions** (CAC Certified Enterprise: Reg 9912717)
> • Admin & Technical Desk Hotline: **0802 279 1227**
> • WhatsApp Support: **wa.me/2348022791227**
> • Corporate Bank Account: **Moniepoint Microfinance Bank (6805375225)**